

Written HW 1 — Full Solutions

Problem statements from Knight, *Physics for Scientists and Engineers* (5th ed., Global Edition), Chapter 15. $g = 9.8 \text{ m/s}^2$ throughout.

15.19 (3 pts)

A 500 g air-track glider moving at 0.50 m/s collides with a horizontal spring whose opposite end is anchored to the end of the track. Measurements show that the glider is in contact with the spring for 1.5 s before it rebounds.

- What is the value of the spring constant?
- What is the maximum compression of the spring?

MODEL: We will model the spring as ideal, and the air track allows us to ignore friction. Although the air glider is not attached to the spring, for the portion of the motion during which the glider is compressing the spring, the motion is exactly the same as if it were attached.

VISUALIZE: We know the period of an oscillation is given by

$$T = 2\pi\sqrt{m/k}$$

The time the glider is in contact with the spring is only one half of a full oscillation. We can use this to solve part (a). For part (b), we will use energy conservation.

SOLVE: (a) We know

$$T/2 = \pi\sqrt{m/k} = t \Rightarrow k = m(\pi/t)^2 = (0.500 \text{ kg})(\pi/1.5 \text{ s})^2 = 2.19 \text{ N/m} \approx \mathbf{2.2 \text{ N/m}}$$

(b) Call the initial condition the moment before the glider strikes the spring, and call the final condition the moment of maximum compression. Energy conservation tells us

$$E_i = E_f \Rightarrow \frac{1}{2}mv^2 = \frac{1}{2}k(\Delta x)^2$$

$$\Rightarrow \Delta x = v\sqrt{m/k} = (0.50 \text{ m/s})\sqrt{(0.500 \text{ kg}) / 2.19 \text{ N/m}} = 0.238 \text{ m} \approx \mathbf{0.24 \text{ m}}$$

REVIEW: This is almost a foot of compression, which seems like a lot. But keep in mind the spring is not very stiff, only exerting a force of 2.2 N (about half a pound) when compressed by a full meter. In light of that, this compression seems reasonable.

15.23 (3 pts)

A spring with spring constant 15 N/m hangs from the ceiling. A ball is attached to the spring and allowed to come to rest. It is then pulled down 6.0 cm and released. If the ball makes 30 oscillations in 20 s, what are its (a) mass and (b) maximum speed?

MODEL: The vertical oscillations constitute simple harmonic motion.

SOLVE: The period and angular frequency are

$$T = 20 \text{ s} / 30 \text{ oscillations} = 0.6667 \text{ s} \quad \text{and} \quad \omega = 2\pi/T = 2\pi/0.6667 \text{ s} = 9.425 \text{ rad/s}$$

(a) The mass can be found as follows:

$$\omega = \sqrt{k/m} \Rightarrow m = k/\omega^2 = (15 \text{ N/m})/(9.425 \text{ rad/s})^2 = 0.169 \text{ kg} \approx \mathbf{0.17 \text{ kg}}$$

(b) The maximum speed

$$v_{\text{max}} = \omega A = (9.425 \text{ rad/s})(0.060 \text{ m}) = \mathbf{0.57 \text{ m/s}}$$

REVIEW: Note that the average speed is four times the amplitude divided by the period, or 24 cm / 0.67 s, which is 36 cm/s. It is reasonable that the maximum speed would be a little larger than this, but of a similar order of magnitude. Our answer to (b) is reasonable.

15.24 (3 pts)

A 200 g ball is tied to a string. It is pulled to an angle of 8.0° and released to swing as a pendulum. A student with a stopwatch finds that 10 oscillations take 12 s. How long is the string?

MODEL: Assume the small-angle approximation so there is simple harmonic motion.

SOLVE: The period is

$$T = 12 \text{ s} / 10 \text{ oscillations} = 1.20 \text{ s}$$

and is given by the formula

$$T = 2\pi\sqrt{L/g} \Rightarrow L = (T/2\pi)^2 g = (1.20 \text{ s} / 2\pi)^2 (9.8 \text{ m/s}^2) = \mathbf{0.36 \text{ m (36 cm)}}$$

Note that neither the 200 g mass nor the 8.0° angle enters this calculation — a pendulum's period depends only on its length (and g), not on its mass or amplitude, as long as the small-angle approximation holds.

15.28 (4 pts, +1 extra credit)

A pendulum on a 75-cm-long string has a maximum speed of 0.25 m/s. What is the pendulum's maximum angle in degrees?

Extra credit: solve this problem using energy conservation. You may assume the mass is 100 g.

MODEL: We will model this as a simple pendulum, which is equivalent to saying we will assume the mass of the string is negligible, and that all mass is concentrated at the bob on the end.

VISUALIZE: The position of the bob along its arc is $s(t) = A \cos(\omega t + \phi_0) = L\theta_{\max} \cos(\omega t + \phi_0)$. Differentiating with respect to time gives the speed:

$$ds/dt = -L\theta_{\max} \omega \sin(\omega t + \phi_0)$$

SOLVE: The maximum speed is clearly $v_{\max} = L\theta_{\max} \omega = \theta_{\max} \sqrt{gL}$. Rearranging,

$$\theta_{\max} = v_{\max} / \sqrt{gL} = (0.25 \text{ m/s}) / \sqrt{[(9.8 \text{ m/s}^2)(0.75 \text{ m})]} = 0.092 \text{ rad or } \mathbf{5.3^\circ}$$

REVIEW: This is a small angle, which justifies our reliance on results from the chapter that used the small-angle approximation. This is consistent!

Extra credit — energy-conservation method:

With $m = 100 \text{ g}$, energy is conserved between the bottom of the swing (all kinetic) and the highest point of the swing (all gravitational potential), where the bob has risen a height $h = L(1 - \cos\theta_{\max})$:

$$\frac{1}{2}mv_{\max}^2 = mgL(1 - \cos\theta_{\max})$$

The mass m cancels from both sides, so it was never actually needed — but carrying it through:

$$\begin{aligned} \cos\theta_{\max} &= 1 - v_{\max}^2 / (2gL) \\ &= 1 - (0.25 \text{ m/s})^2 / [2(9.8 \text{ m/s}^2)(0.75 \text{ m})] = 1 - 0.00425 = 0.99575 \end{aligned}$$

$$\Rightarrow \theta_{\max} = \cos^{-1}(0.99575) = \mathbf{5.3^\circ}$$

The two methods agree exactly, as they must — the small-angle SHM approach and direct energy conservation describe the same physics.

15.49 (5 pts)

A 200 g block hangs from a spring with spring constant 10 N/m. At $t = 0$ s the block is 20 cm below the equilibrium point and moving upward with a speed of 100 cm/s. What are the block's

- Oscillation frequency?
- Distance from equilibrium when the speed is 50 cm/s?
- Distance from equilibrium at $t = 1.0$ s?

MODEL: The block undergoes simple harmonic motion.

SOLVE: (a) The frequency of oscillation is

$$f = (1/2\pi)\sqrt{(k/m)} = (1/2\pi)\sqrt{(10 \text{ N/m} / 0.20 \text{ kg})} = 1.125 \text{ Hz} \approx \mathbf{1.1 \text{ Hz}}$$

(b) Using conservation of energy, $\frac{1}{2}mv_1^2 + \frac{1}{2}kx_1^2 = \frac{1}{2}mv_0^2 + \frac{1}{2}kx_0^2$, we find

$$\begin{aligned} x_1 &= \sqrt{[x_0^2 + (m/k)(v_0^2 - v_1^2)]} \\ &= \sqrt{[(-0.20 \text{ m})^2 + (0.20 \text{ kg}/10 \text{ N/m})((1.00 \text{ m/s})^2 - (0.50 \text{ m/s})^2)]} \\ &= \mathbf{0.2345 \text{ m, or } 23 \text{ cm}} \end{aligned}$$

(c) At time t , the displacement is $x = A \cos(\omega t + \phi_0)$. The angular frequency is $\omega = 2\pi f = 7.071 \text{ rad/s}$. The amplitude is $A = \sqrt{[x_0^2 + (v_0/\omega)^2]} = \sqrt{[(-0.20 \text{ m})^2 + (1.00 \text{ m/s} / 7.071 \text{ rad/s})^2]} = 0.245 \text{ m}$

The phase constant is

$$\phi_0 = \cos^{-1}(x_0/A) = \cos^{-1}(-0.200 \text{ m} / 0.245 \text{ m}) = \pm 2.526 \text{ rad or } \pm 145^\circ$$

A negative displacement (below the equilibrium point) and positive velocity (upward motion) indicate that the corresponding circular motion is in the third quadrant, so $\phi_0 = -2.526 \text{ rad}$. Thus at $t = 1.0$ s,

$$x = (0.245 \text{ m}) \cos[(7.071 \text{ rad/s})(1.0 \text{ s}) - 2.526 \text{ rad}] = -0.0409 \text{ m} = \mathbf{-4.1 \text{ cm}}$$

The block is 4.1 cm below the equilibrium point.